Omoze Idehenre
3/22/2026
PKI Artifacts Week 3

Pems

```
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ ls labs/week-03/submiss
ions/*.pem
labs/week-03/submissions/github_cert.pem
labs/week-03/submissions/intermediate.pem
labs/week-03/submissions/leaf_cert.pem
labs/week-03/submissions/server.pem
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```

Google Certificate

```
_cert.pem -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            aa:23:02:42:8e:f4:39:7e:10:bb:2c:32:93:1c:fc:2e
    Signature Algorithm: ecdsa-with-SHA256
        Issuer: C=US, O=Google Trust Services, CN=WE2
        Validity
            Not Before: Feb 23 18:19:56 2026 GMT
            Not After : May 18 18:19:55 2026 GMT
        Subject: CN=*.google.com
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:c5:be:e6:02:1e:38:8d:f8:fa:51:b7:ec:cc:06:
                    b1:08:3f:86:f8:76:a2:8d:67:74:9f:c6:1d:d6:b0:
                    2c:88:2a:5d:36:0d:cc:a0:4e:54:bf:66:70:f5:71:
                    46:a1:4f:c3:5b:c9:43:45:d6:e6:99:07:89:34:d6:
                    2a:47:c0:65:90
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Key Usage: critical
                Digital Signature
            X509v3 Extended Key Usage:
                TLS Web Server Authentication
            X509v3 Basic Constraints: critical
                CA:FALSE
            X509v3 Subject Key Identifier:
                E1:9F:20:4A:4B:B8:79:A3:1F:A8:5F:A0:D5:73:86:9E:00:9F:A1:23
            X509v3 Authority Key Identifier:
```

Extensions

```
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Key Usage: critical
                Digital Signature
            X509v3 Extended Key Usage:
                TLS Web Server Authentication
            X509v3 Basic Constraints: critical
                CA:FALSE
            X509v3 Subject Key Identifier:
                E1:9F:20:4A:4B:B8:79:A3:1F:A8:5F:A0:D5:73:86:9E:00:9F:A1:23
            X509v3 Authority Key Identifier:
                keyid:75:BE:C4:77:AE:89:F6:44:37:7D:CF:B1:68:1F:1D:1A:EB:DC:34:59

            Authority Information Access:
                OCSP - URI:http://o.pki.goog/we2
                CA Issuers - URI:http://i.pki.goog/we2.crt

            X509v3 Subject Alternative Name:
                DNS:*.google.com, DNS:*.appengine.google.com, DNS:*.bdn.dev, DNS:*.ori
gin-test.bdn.dev, DNS:*.cloud.google.com, DNS:*.crowdsource.google.com, DNS:*.datacomp
ute.google.com, DNS:*.google.ca, DNS:*.google.cl, DNS:*.google.co.in, DNS:*.google.co.
jp, DNS:*.google.co.uk, DNS:*.google.com.ar, DNS:*.google.com.au, DNS:*.google.com.br,
 DNS:*.google.com.co, DNS:*.google.com.mx, DNS:*.google.com.tr, DNS:*.google.com.vn, D
NS:*.google.de, DNS:*.google.es, DNS:*.google.fr, DNS:*.google.hu, DNS:*.google.it, DN
S:*.google.nl, DNS:*.google.pl, DNS:*.google.pt, DNS:*.googleapis.cn, DNS:*.gstatic.cn
, DNS:*.gstatic-cn.com, DNS:googlecnapps.cn, DNS:*.googlecnapps.cn, DNS:googleapps-cn.
com, DNS:*.googleapps-cn.com, DNS:gkecnapps.cn, DNS:*.gkecnapps.cn, DNS:googledownload
s.cn, DNS:*.googledownloads.cn, DNS:recaptcha.net.cn, DNS:*.recaptcha.net.cn, DNS:reca
ptcha-cn.net, DNS:*.recaptcha-cn.net, DNS:widevine.cn, DNS:*.widevine.cn, DNS:ampproje
ct.org.cn, DNS:*.ampproject.org.cn, DNS:ampproject.net.cn, DNS:*.ampproject.net.cn, DN
S:google-analytics-cn.com, DNS:*.google-analytics-cn.com, DNS:googleadservices-cn.com,
 DNS:*.googleadservices-cn.com, DNS:googlevads-cn.com, DNS:*.googlevads-cn.com, DNS:go
```

Leaf Certificate

```
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in labs/week-03/submissions/github_cert.pem -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            1d:c2:89:c1:ea:da:fb:04:e9:d1:cf:53:d5:d7:22:53
    Signature Algorithm: ecdsa-with-SHA256
        Issuer: C=GB, O=Sectigo Limited, CN=Sectigo Public Server Authentication CA DV E36
        Validity
            Not Before: Mar  6 00:00:00 2026 GMT
            Not After : Jun  3 23:59:59 2026 GMT
        Subject: CN=github.com
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:3d:4a:7d:f8:be:fb:3d:e4:6c:73:2c:c5:03:e7:
                    17:eb:76:46:9e:45:c6:d5:6c:b2:41:2a:32:c8:9a:
                    e5:79:eb:5a:48:4d:05:c9:42:2d:b2:63:6b:7a:c2:
                    75:c4:19:49:e9:7a:27:33:07:f7:08:26:65:9f:12:
                    7c:38:2e:56:5a
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Authority Key Identifier:
                keyid:17:99:A8:04:C1:6F:E4:2D:70:A8:0A:10:3D:03:D3:E9:1A:B8:26:63

            X509v3 Subject Key Identifier:
                08:13:82:D1:04:B9:7D:00:6B:A2:B1:81:22:83:C7:C7:DC:4F:8C:DC
            X509v3 Key Usage: critical
```

```
            X509v3 Subject Key Identifier:
                08:13:82:D1:04:B9:7D:00:6B:A2:B1:81:22:83:C7:C7:DC:4F:8C:DC
            X509v3 Key Usage: critical
                Digital Signature
            X509v3 Basic Constraints: critical
                CA:FALSE
            X509v3 Extended Key Usage:
                TLS Web Server Authentication
            X509v3 Certificate Policies:
                Policy: 1.3.6.1.4.1.6449.1.2.2.7
                  CPS: https://sectigo.com/CPS
                Policy: 2.23.140.1.2.1

            Authority Information Access:
                CA Issuers - URI:http://crt.sectigo.com/SectigoPublicServerAuthenticat
ionCADVE36.crt
                OCSP - URI:http://ocsp.sectigo.com

            1.3.6.1.4.1.11129.2.4.2:
                ......v..W.....>3.,........=q2%.!.%.a.N!.............G0E. ....2..}.e..
.!.8n?o...........Y..!..O.w..z.g.p..F.f74..c[B(...y..f..u...-...%...:.E....#..K..)'...
3................F0D. .<..@>8...b...#J"..la
....6be=8_. -..?.N..Y.9>.s.+ ;...
            X509v3 Subject Alternative Name:
                DNS:github.com, DNS:www.github.com
    Signature Algorithm: ecdsa-with-SHA256
         30:44:02:20:2c:c4:c3:aa:0d:9f:fb:e9:7b:ef:9c:35:8c:1c:
         bd:95:2c:0b:9f:f7:35:82:4f:db:03:7c:61:81:5a:3e:b1:0f:
         02:20:58:a4:52:31:e2:50:18:8e:90:f2:42:c7:a4:b3:46:02:
         15:ca:73:47:27:4b:2f:65:3f:17:99:e7:e1:5e:f4:b0
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```

Intermediate Certificate

```
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in labs/w
eek-03/submissions/github_cert.pem -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            1d:c2:89:c1:ea:da:fb:04:e9:d1:cf:53:d5:d7:22:53
    Signature Algorithm: ecdsa-with-SHA256
        Issuer: C=GB, O=Sectigo Limited, CN=Sectigo Public Server Authentication CA DV
 E36
        Validity
            Not Before: Mar  6 00:00:00 2026 GMT
            Not After : Jun  3 23:59:59 2026 GMT
        Subject: CN=github.com
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:3d:4a:7d:f8:be:fb:3d:e4:6c:73:2c:c5:03:e7:
                    17:eb:76:46:9e:45:c6:d5:6c:b2:41:2a:32:c8:9a:
                    e5:79:eb:5a:48:4d:05:c9:42:2d:b2:63:6b:7a:c2:
                    75:c4:19:49:e9:7a:27:33:07:f7:08:26:65:9f:12:
                    7c:38:2e:56:5a
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Authority Key Identifier:
                keyid:17:99:A8:04:C1:6F:E4:2D:70:A8:0A:10:3D:03:D3:E9:1A:B8:26:63

            X509v3 Subject Key Identifier:
                08:13:82:D1:04:B9:7D:00:6B:A2:B1:81:22:83:C7:C7:DC:4F:8C:DC
            X509v3 Key Usage: critical
                Digital Signature
```

```
                Digital Signature
            X509v3 Basic Constraints: critical
                CA:FALSE
            X509v3 Extended Key Usage:
                TLS Web Server Authentication
            X509v3 Certificate Policies:
                Policy: 1.3.6.1.4.1.6449.1.2.2.7
                  CPS: https://sectigo.com/CPS
                Policy: 2.23.140.1.2.1

            Authority Information Access:
                CA Issuers - URI:http://crt.sectigo.com/SectigoPublicServerAuthenticat
ionCADVE36.crt
                OCSP - URI:http://ocsp.sectigo.com

            1.3.6.1.4.1.11129.2.4.2:
                ......v..W.....>3.,........=q2%.!.%.a.N!.............G0E. ....2..}.e..
.!.8n?o...........Y..!..O.w..z.g.p..F.f74..c[B(...y..f..u...-...%...:.E....#..K..)'...
3................F0D. .<..@>8...b...#J"..la
....6be=8_. -..?.N..Y.9>.s.+ ;...
            X509v3 Subject Alternative Name:
                DNS:github.com, DNS:www.github.com
    Signature Algorithm: ecdsa-with-SHA256
         30:44:02:20:2c:c4:c3:aa:0d:9f:fb:e9:7b:ef:9c:35:8c:1c:
         bd:95:2c:0b:9f:f7:35:82:4f:db:03:7c:61:81:5a:3e:b1:0f:
         02:20:58:a4:52:31:e2:50:18:8e:90:f2:42:c7:a4:b3:46:02:
         15:ca:73:47:27:4b:2f:65:3f:17:99:e7:e1:5e:f4:b0
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$ openssl x509 -in labs/w
eek-03/submissions/intermediate.pem -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
```

```
        Serial Number:
            36:e0:59:ed:88:8d:d5:7a:ed:d5:70:b6:72:7f:9b:da
    Signature Algorithm: ecdsa-with-SHA384
        Issuer: C=GB, O=Sectigo Limited, CN=Sectigo Public Server Authentication Root
E46
        Validity
            Not Before: Mar 22 00:00:00 2021 GMT
            Not After : Mar 21 23:59:59 2036 GMT
        Subject: C=GB, O=Sectigo Limited, CN=Sectigo Public Server Authentication CA D
V E36
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:68:a1:a7:6c:05:27:05:89:63:1c:39:a7:ff:25:
                    21:c5:ed:d3:2f:12:98:bb:2c:dc:f5:55:e8:49:a0:
                    84:57:91:7b:dc:58:5f:6b:f4:a2:fd:13:2c:9b:04:
                    c5:5b:74:7d:b3:b2:7a:26:96:19:16:6b:b8:76:8d:
                    ef:93:f5:60:97
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Authority Key Identifier:
                keyid:D1:22:DA:4C:59:F1:4B:5F:26:38:AA:9D:D6:EE:EB:0D:C3:FB:A9:61

            X509v3 Subject Key Identifier:
                17:99:A8:04:C1:6F:E4:2D:70:A8:0A:10:3D:03:D3:E9:1A:B8:26:63
            X509v3 Key Usage: critical
                Digital Signature, Certificate Sign, CRL Sign
            X509v3 Basic Constraints: critical
                CA:TRUE, pathlen:0
            X509v3 Extended Key Usage:
                TLS Web Server Authentication, TLS Web Client Authentication
```

```
            X509v3 Extended Key Usage:
                TLS Web Server Authentication, TLS Web Client Authentication
            X509v3 Certificate Policies:
                Policy: X509v3 Any Policy
                Policy: 2.23.140.1.2.1

            X509v3 CRL Distribution Points:

                Full Name:
                  URI:http://crl.sectigo.com/SectigoPublicServerAuthenticationRootE46.
crl

            Authority Information Access:
                CA Issuers - URI:http://crt.sectigo.com/SectigoPublicServerAuthenticat
ionRootE46.p7c
                OCSP - URI:http://ocsp.sectigo.com

    Signature Algorithm: ecdsa-with-SHA384
         30:64:02:30:5b:0a:9c:14:03:87:ae:25:fa:ff:9a:51:85:a3:
         0c:22:4a:40:cc:47:51:41:a6:73:00:18:0c:88:c9:f6:96:d1:
         60:8b:4c:08:8c:79:c6:ed:28:51:be:ac:21:f4:d7:a9:02:30:
         3d:d7:5c:b0:31:38:29:d7:8b:bb:58:49:76:91:e0:74:bb:ec:
         1c:02:b7:bf:28:32:2c:6b:a3:31:4f:71:04:90:83:69:c9:11:
         4f:dd:d5:1c:41:34:82:9a:60:36:a7:55
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```

```
        Serial Number:
            36:e0:59:ed:88:8d:d5:7a:ed:d5:70:b6:72:7f:9b:da
    Signature Algorithm: ecdsa-with-SHA384
        Issuer: C=GB, O=Sectigo Limited, CN=Sectigo Public Server Authentication Root
E46
        Validity
            Not Before: Mar 22 00:00:00 2021 GMT
            Not After : Mar 21 23:59:59 2036 GMT
        Subject: C=GB, O=Sectigo Limited, CN=Sectigo Public Server Authentication CA D
V E36
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:68:a1:a7:6c:05:27:05:89:63:1c:39:a7:ff:25:
                    21:c5:ed:d3:2f:12:98:bb:2c:dc:f5:55:e8:49:a0:
                    84:57:91:7b:dc:58:5f:6b:f4:a2:fd:13:2c:9b:04:
                    c5:5b:74:7d:b3:b2:7a:26:96:19:16:6b:b8:76:8d:
                    ef:93:f5:60:97
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Authority Key Identifier:
                keyid:D1:22:DA:4C:59:F1:4B:5F:26:38:AA:9D:D6:EE:EB:0D:C3:FB:A9:61

            X509v3 Subject Key Identifier:
                17:99:A8:04:C1:6F:E4:2D:70:A8:0A:10:3D:03:D3:E9:1A:B8:26:63
            X509v3 Key Usage: critical
                Digital Signature, Certificate Sign, CRL Sign
            X509v3 Basic Constraints: critical
                CA:TRUE, pathlen:0
            X509v3 Extended Key Usage:
                TLS Web Server Authentication, TLS Web Client Authentication
```

```
            X509v3 Extended Key Usage:
                TLS Web Server Authentication, TLS Web Client Authentication
            X509v3 Certificate Policies:
                Policy: X509v3 Any Policy
                Policy: 2.23.140.1.2.1

            X509v3 CRL Distribution Points:

                Full Name:
                  URI:http://crl.sectigo.com/SectigoPublicServerAuthenticationRootE46.
crl

            Authority Information Access:
                CA Issuers - URI:http://crt.sectigo.com/SectigoPublicServerAuthenticat
ionRootE46.p7c
                OCSP - URI:http://ocsp.sectigo.com

    Signature Algorithm: ecdsa-with-SHA384
         30:64:02:30:5b:0a:9c:14:03:87:ae:25:fa:ff:9a:51:85:a3:
         0c:22:4a:40:cc:47:51:41:a6:73:00:18:0c:88:c9:f6:96:d1:
         60:8b:4c:08:8c:79:c6:ed:28:51:be:ac:21:f4:d7:a9:02:30:
         3d:d7:5c:b0:31:38:29:d7:8b:bb:58:49:76:91:e0:74:bb:ec:
         1c:02:b7:bf:28:32:2c:6b:a3:31:4f:71:04:90:83:69:c9:11:
         4f:dd:d5:1c:41:34:82:9a:60:36:a7:55
Omozes-MacBook-Pro:PKI-Portfolio-OmozeIdehenre- omozeidehenre$
```